Test rock hardness and water

Today I can… □ show a safe mark test □ watch where drops go □ tell what I saw

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Staff answers

Starter

1. Check a property you can see or feel. 2. Use it again if safe. 3. A skeleton or bones. Confirm actual taught coverage.

I do

A wet top is not proof that water passed through. Check the underside and tray, and check that water did not run round the edge. The model shows possible routes, not results.

We do

Use an adult-held blunt tool: Mark test. Add three drops: Water test. Look in the tray below: Water test. Hand over the missing observation at B; do not supply a guessed result.

We do 2

Step check: Drops in the tray below is the intended choice; the route must also be checked. Hinge: Dry the tray, then repeat the test.

Option	Diagnosis
Call C porous from the wet tray.	Uses water present before the test as a result.
Call C hard from the wet tray.	Confuses water with hardness.
Skip the test and guess from colour.	Uses appearance rather than the planned check.
Dry the tray, then repeat the test.	Removes old water before observing a new result.

About 3 in 4 or more correct: move on. Otherwise re-model the source example and ask the parallel hinge. For a small group, check each learner’s explanation rather than infer security from a percentage.

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Staff answers · check and adapt

Parallel hinge

A drop from new rock D runs round its edge into a dry tray. Does that prove water passed through D? No. Check the route and repeat safely.

Independent and Exit

Accept only what the learner observed. Mark and water are separate tests. A hard sample may also allow water through. Do not infer a water result from hardness or a rock name. Run-off: reset and inspect a stable sample and dry tray; repeat safely with the same drops. If the route is unclear, report that limit.

Practical limit

This open tray does not by itself establish rock permeability. Inspect whether drops passed through the sample or round its edge. Preserve “Water route seen” as the results heading. Do not claim permeability from tray drops alone.

Safety

Use checked clean blunt samples, a stable tray and towel. Staff use the blunt tool, inspect edges and control water. No dust, tasting or forced touch. Wash hands. Check the school risk assessment and [SYS] guidance.

Misconceptions

A wet top proves water went through: compare top, underside, edge and tray. Scratch and water tests measure the same thing: keep two result columns and use the separate observation for each.

Fading

Full model: Adult uses one blunt tool. A gets a mark. Three drops stay on top. Shared model: Mark B. Add three drops. The drops ____. Independent: no filled observation. Next fade: learner picks the step and result from unlabelled kit.

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Staff guidance · access and evidence

Adaptations

Cognition and learning: offer a three-step visual card and remove the last cue after a correct try. Communication: point to real kit before a short spoken, signed or AAC response and wait. SEMH: private first–then cue and planned pause; re-enter with one step. Sensory or physical: stable tray, large cards or accessible input, with no forced touch. EAL: rehearse an Independent or Aspire idea in any language before recording; retain the science challenge.

Evidence and feedback

Record the learner's observation or results table, review edit and exit response. Use SCI-BUILD-A2-W03 beside written, spoken, practical or photographic evidence. Record adult prompts separately. At We do 2 record WS or I and the actual prompt level. At Review give VF and a specific NS(+). Exit may include self-edit // or read back ?. R requires a response to a next step heard by an adult; never award it automatically. Capture E on [SYS] only for an observable curriculum action and never while distressed.

If behind

Keep both safe tests and the learner's own results. After a secure hinge, shorten repeated oral rehearsal. If needed, merge exit reflection into the Week 4 Arrival and record reason and coverage impact. Pass on actual understanding and unresolved hinge errors. Do not replace individual work with a group product.

Individual opportunities

Record each learner's own sequencing, attention, communication, turn-taking, re-entry or access target and prompt level. EAL is a language need, not SEN. Follow the learner's [SYS] where it differs from these routines. A lab technician compares rock samples with the same checks.

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